

Date: 07/10/2025

Submission Date: 13/10/2025 (11:55 PM)

Note: Below is the list of questions on **MOS Transistors** and **Small-Signal Parameters of a MOS Transistor**. You are required to:

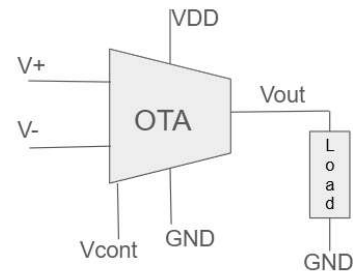
- Perform the simulations using **Cadence Virtuoso** with **SCL 180 nm PDK**.
- Prepare a detailed report based on your work.
- Submit the report for evaluation.

Report Guidelines:

- Include a **Conclusion** section at the end, briefly summarizing your insights and observations from the design process.
 - Name the report file as: RollNumber_Assignment4.pdf
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1. Design an Operational **Transconductance Amplifier (OTA)** for the following specifications.

- Differential Gain = 40 - 60 dB (Gain should vary between 40 to 60 dB by changing Vcont)
- CMRR > 70dB
- ICMR: 0.80 V – 1.4 V
- Output Swing > 800m V
- Power < 20 uW
- Load Capacitance = 1pF



- Perform frequency analysis and calculate bandwidth of the designed amplifier. Plot the frequency response (both Gain and Phase) with the capacitive load.
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Report should include,

- Transistor level circuit diagram (full system and individual sub-blocks) and sizing (with overdrive voltage, gm, gds and region) table of all transistors.
- Step-by-step design methodology/derivations/calculations (in brief).
- Overall power consumption and its block-wise breakup should be provided.
- Plots should be neatly labelled, and labels should be clearly visible
- Final results should be provided in tabular format
- Any other information as deemed necessary